

Task 1

1. It will add padding.
2. It takes significantly longer (~2 seconds for the intro, over 18 for this)

```
gio@Computery: ~/Documen X + v
$ md5collgen -p q2.txt -o q2out1.bin q2out2.bin
MD5 collision generator v1.5
by Marc Stevens (http://www.win.tue.nl/hashclash/)

Using output filenames: 'q2out1.bin' and 'q2out2.bin'
Using prefixfile: 'q2.txt'
Using initial value: 96c45fb804e3410aae10ed603e8e763e

Generating first block: .....
Generating second block: S10.....
Running time: 18.3871 s

(gio@Computery) ~/Documents/school/cse4400/lab1
$ ll
total 3288
-rwxr-xr-x 1 gio gio 3338360 Mar 23 2021 md5collgen
-rw-r--r-- 1 gio gio 192 Feb 22 21:50 out1.bin
-rw-r--r-- 1 gio gio 192 Feb 22 21:50 out2.bin
-rw-r--r-- 1 gio gio 53 Feb 22 21:49 prefix.txt
-rw-r--r-- 1 gio gio 192 Feb 22 22:06 q2out1.bin
-rw-r--r-- 1 gio gio 192 Feb 22 22:06 q2out2.bin
-rw-r--r-- 1 gio gio 64 Feb 22 22:00 q2.txt

(gio@Computery) ~/Documents/school/cse4400/lab1
$ md5sum q2out1.bin
8520f17b0e61857b75cb88c7438f3c03 q2out1.bin

(gio@Computery) ~/Documents/school/cse4400/lab1
$ md5sum q2out2.bin
8520f17b0e61857b75cb88c7438f3c03 q2out2.bin
```

3.

53	69	6E	63	65	20	49	20	77	61	73	20	62	6F	72	6E	2C	20	49	27	76	65	20	68
65	61	72	64	20	74	68	65	20	76	6F	69	63	65	20	6F	66	20	74	68	65	20	66	61
74	68	65	72	0A	47	6F	65	73	20	69	6E	20	74	20	0A	12	2D	51	F8	38	70	00	AA
A7	C0	B9	C9	2A	A2	6A	0B	05	36	47	B8	0B	F6	25	C2	38	A6	83	8A	29	AE	9C	04
D2	01	BA	CB	7A	EC	0D	78	0B	6C	4A	53	C2	8B	76	56	70	E6	1F	AD	85	17	3E	4B
9E	A3	06	91	7D	33	C5	12	3B	E3	85	F1	41	33	52	3C	39	03	D1	1E	A6	5D	4E	14
27	B4	04	44	38	9D	72	FF	8B	C5	00	CF	8B	5F	48	B7	D7	75	58	48	97	06	2D	FC
FA	46	46	7B	74	94	0A	9F	D8	F1	9E	7B	8E	A2	64	E7	5F	46	E2	A2	76	14	FE	15

53	69	6E	63	65	20	49	20	77	61	73	20	62	6F	72	6E	2C	20	49	27	76	65	20	68
65	61	72	64	20	74	68	65	20	76	6F	69	63	65	20	6F	66	20	74	68	65	20	66	61
74	68	65	72	0A	47	6F	65	73	20	69	6E	20	74	20	0A	12	2D	51	F8	38	70	00	AA
A7	C0	B9	C9	2A	A2	6A	0B	05	36	47	38	0B	F6	25	C2	38	A6	83	8A	29	AE	9C	04
D2	01	BA	CB	7A	EC	0D	78	0B	6C	4A	53	C2	0B	77	56	70	E6	1F	AD	85	17	3E	4B
9E	A3	06	11	7D	33	C5	12	3B	E3	85	F1	41	33	52	3C	39	03	D1	1E	A6	5D	4E	14
27	B4	04	C4	38	9D	72	FF	8B	C5	00	CF	8B	5F	48	B7	D7	75	58	48	97	06	2D	FC
FA	46	46	7B	74	14	0A	9F	D8	F1	9E	7B	8E	A2	64	E7	5F	46	E2	22	76	14	FE	15

The different bytes are circled.

Task 3

```
(gio@Computery)~[/Documents/school/cse4400/lab1/task3]
$ cat task3out1.bin suffix > coll1

(gio@Computery)~[/Documents/school/cse4400/lab1/task3]
$ cat task3out2.bin suffix > coll2

(gio@Computery)~[/Documents/school/cse4400/lab1/task3]
$ md5sum coll1
d253a30d7f0da3e5b09130be27d6a344 coll1

(gio@Computery)~[/Documents/school/cse4400/lab1/task3]
$ md5sum coll2
d253a30d7f0da3e5b09130be27d6a344 coll2

(gio@Computery)~[/Documents/school/cse4400/lab1/task3]
$ diff coll1 coll2
Binary files coll1 and coll2 differ
```

Task 4

```
$ ll
total 144
-rwxr-xr-x 1 gio gio 16472 Feb 22 23:25 coll1.o
-rwxr-xr-x 1 gio gio 16472 Feb 22 23:25 coll2.o
-rw-r--r-- 1 gio gio 12352 Feb 22 23:03 prefix
-rw-r--r-- 1 gio gio 3992 Feb 22 23:25 suffix
-rw-r--r-- 1 gio gio 3993 Feb 22 23:05 suffix.bak
-rw-r--r-- 1 gio gio 2669 Feb 22 22:52 task4.c
-rw-r--r-- 1 gio gio 2669 Feb 22 23:09 task4coll.c
-rwxr-xr-x 1 gio gio 16472 Feb 22 23:09 task4coll.o
-rwxr-xr-x 1 gio gio 16472 Feb 22 22:52 task4.o
-rw-r--r-- 1 gio gio 12480 Feb 22 23:03 task4out1.bin
-rw-r--r-- 1 gio gio 12480 Feb 22 23:03 task4out2.bin

(gio@Computery)~[/Documents/school/cse4400/lab1/task4]
$ md5sum coll1.o
42d17cdf4a4ee29af78497c1dd704f0d coll1.o

(gio@Computery)~[/Documents/school/cse4400/lab1/task4]
$ md5sum coll2.o
42d17cdf4a4ee29af78497c1dd704f0d coll2.o

(gio@Computery)~[/Documents/school/cse4400/lab1/task4]
$ ./coll1.o
safe

(gio@Computery)~[/Documents/school/cse4400/lab1/task4]
$ ./coll2.o
pwned
```